ECON 4555 — Part III, Question 6

Modern Table 2: Fare Regressions

2025 Q2 — Reduced-control OLS specification

	Ordinary least squares		
Dependent variable	$\ln(P_{20})$	$\ln(P_{50})$	$\ln(P_{80})$
ln(Distance)	0.2497***	0.2168***	0.2292***
	(0.0064)	(0.0060)	(0.0055)
ln(LoadFactor)	0.3264***	0.3864***	0.2601***
	(0.0453)	(0.0456)	(0.0381)
ln(Circuity)	0.1912***	0.1601***	0.1385***
	(0.0238)	(0.0214)	(0.0193)
Coupon-stop proxy	0.1140***	0.1037***	0.0432***
	(0.0083)	(0.0072)	(0.0068)
Airline airport-origin share	0.2505***	0.2947***	0.2859***
	(0.0299)	(0.0274)	(0.0259)
Airport-origin HHI	-0.2332***	-0.2256***	-0.2224***
	(0.0325)	(0.0291)	(0.0273)
Airline route share	0.1467***	0.0898***	0.0516***
	(0.0196)	(0.0171)	(0.0160)
Route HHI	0.0637***	0.0602***	0.0884***
	(0.0246)	(0.0217)	(0.0201)
Carrier fixed effects	Yes	Yes	Yes
Constant	Yes	Yes	Yes
Observations	7,499	7,499	7,499
R^2	0.8485	0.8731	0.8740
Adjusted R^2	0.8481	0.8727	0.8737

Notes. HC1 heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; two-sided tests use a t reference distribution with 7,477 residual degrees of freedom. All eight reported slope coefficients are significant at the 1% level in each model under this inference convention. Carrier coefficients and the intercept are estimated but omitted from display. R^2 includes carrier fixed effects, not a within-carrier-only fit measure.

Estimation sample. Unweighted OLS on the unchanged Part II baseline: one reproducibly selected eligible ticketing carrier for each of 7,499 unordered airport pairs. All modeled variables are valid in all rows; no further observations are excluded. Eligibility requires at least 10 estimated passengers/day and a route share of at least 10%, with the inherited cleaning rules. There are 14 carrier categories (13 dummies plus an intercept), eight slope controls and 22 estimated parameters. The reference carrier is 3M; its choice does not change the reported slopes.

Variables and comparison. Fares are passenger-weighted within-carrier/route 20th, 50th and 80th percentiles of allocated directional Market fares; these are three OLS regressions on percentile outcomes, not individual-passenger quantile regressions. Distance, load factor and circuity use natural logs. Shares and HHIs enter in levels on a 0–1 scale. The coupon-stop proxy is a passenger-weighted mean of coupons minus one, not true on-plane stops. Only carrier fixed effects are included. Relative to Borenstein's Table 2, aircraft size, frequency, separate plane changes, competitors' costs, tourism and airport-scarcity controls are omitted; on-plane stops are replaced by the proxy. Optional potential competition and all 2SLS columns are outside this specification.

Reference: Borenstein (1989), Table 2, printed p. 357 / PDF p. 15, especially its final three OLS columns. Sources: validated 2025 Q2 DB1B/T-100 baseline. This is a modern reduced-control comparison, not an exact replication.

Interpreting every slope coefficient

All interpretations condition on the other regressors and carrier fixed effects. Values below are ordered by the 20th-, 50th- and 80th-percentile fare equations. For logged regressors, the coefficient is an elasticity: a 1% relative increase is associated with approximately β% higher fare. A one-percentage-point change in load factor is *not* a 1% relative change.

Illustrative fitted-fare differences. Exact transformations of the estimated log-fare coefficients; they are conditional associations, not intervention effects.

Regressor	Illustrative increase	P_{20}	P_{50}	P_{80}
ln(Distance)	10% relative	+2.41%	+2.09%	+2.21%
ln(LoadFactor)	10% relative	+3.16%	+3.75%	+2.51%
ln(Circuity)	10% relative	+1.84%	+1.54%	+1.33%
Coupon-stop proxy	1 proxy unit	+12.07%	+10.92%	+4.41%
Airline airport-origin share	10 percentage points	+2.54%	+2.99%	+2.90%
Airport-origin HHI	0.10 HHI units	-2.30%	-2.23%	-2.20%
Airline route share	10 percentage points	+1.48%	+0.90%	+0.52%
Route HHI	0.10 HHI units	+0.64%	+0.60%	+0.89%

For log regressors, the exact 10% comparison is $100[(1.10)^{\beta} - 1]$; for a level-regressor change Δx, it is $100[\exp(\beta\Delta x) - 1]$. An HHI increase of 0.10 is 1,000 points on the alternative 0–10,000 scale. These changes describe the exponentiated fitted-log relationship, not an unbiased arithmetic-mean dollar-price prediction.

1. **Distance.** The elasticities are 0.2497, 0.2168, 0.2292. Longer routes have higher total fares, but less than proportionately to distance. Conditional fare per mile therefore declines with distance in this specification.
2. **Load factor.** The elasticities are 0.3264, 0.3864, 0.2601. More heavily occupied service is associated with higher fares. This positive sign does not establish that fuller aircraft cause higher prices or lower efficiency; demand and pricing jointly affect occupancy.
3. **Circuity.** The elasticities are 0.1912, 0.1601, 0.1385. More flown mileage relative to nonstop O&D distance is associated with higher fares after controlling for endpoint distance and the coupon-stop proxy. This is not evidence that passengers value detours.
4. **Coupon-stop proxy.** The coefficients are 0.1140, 0.1037, 0.0432. Moving from zero to one average intermediate coupon point is associated with +12.07%, +10.92%, +4.41% fare differences. The proxy mixes journey structures and cannot distinguish staying on the plane from changing planes; it cannot be read as a price for an additional true stop.
5. **Airline airport-origin share.** The coefficients are 0.2505, 0.2947, 0.2859. A 10-percentage-point increase in the carrier's weighted endpoint origination share is associated with +2.54%, +2.99%, +2.90% higher fares, holding endpoint concentration and route competition measures fixed. Own airport presence matters separately from overall HHI.
6. **Airport-origin HHI.** The coefficients are -0.2332, -0.2256, -0.2224. A 0.10 increase in weighted endpoint HHI is associated with -2.30%, -2.23%, -2.20% fare differences, holding the airline's own airport share fixed. This negative partial association does not imply that airport mergers or less competition would lower fares.
7. **Airline route share.** The coefficients are 0.1467, 0.0898, 0.0516. A 10-percentage-point increase in the airline's route passenger share is associated with +1.48%, +0.90%, +0.52% higher fares, holding route HHI and airport-origin measures fixed. The numerical association is largest for low-percentile fares; no cross-equation equality test is claimed.
8. **Route HHI.** The coefficients are 0.0637, 0.0602, 0.0884. A 0.10 increase in route concentration is associated with +0.64%, +0.60%, +0.89% higher fares at a given own route share. This describes overall concentration conditional on dominance; it is not a forecast of any particular merger.

Competition, methodology and limitations

What the estimates suggest. Higher airline-specific shares on the route and at the endpoint airports are associated with higher fares even after including carrier fixed effects and overall concentration. This is consistent with a role for carrier-specific market position or advantages. Route concentration is positively associated with fares, while airport concentration is negatively associated conditional on own airport share. The results do not support a blanket rule that every form of concentration raises prices. The distinction between a carrier's own dominance and total concentration is essential.

The models' R^2 values are 0.8485, 0.8731 and 0.8740. These summarize in-sample variation in logged fare percentiles explained by the complete specification, including carrier effects. High fit and small p-values do not establish identification, causality or policy reliability. Numerical differences between percentile equations are not formal evidence that coefficients differ statistically across fare percentiles.

Endogeneity. Fares, route shares and airport presence are jointly determined: price affects demand and observed shares, while market position can affect pricing. Load factor, routing and service choices also respond to demand, costs and prices. Unobserved local demand, business-travel intensity, service quality, airport conditions and network structure can correlate with included regressors. Carrier fixed effects remove average carrier-level differences but do not remove route/airport demand shocks. Thus OLS estimates are conditional associations, not causal effects of competition, mergers or service changes. They do not directly measure markups or welfare.

Differences from the original Table 2.

- Only the three OLS equations are recreated. Borenstein also reported instrumental-variable/2SLS specifications addressing endogeneity. No instruments or 2SLS estimates are constructed here. The modern sample has 7,499 routes rather than the original 3,591 observations.
- The original controls include aircraft size, frequency, separate on-plane stops and plane changes, competitors' standardized costs, tourism, and airport scarcity. These are absent from the requested reduced-control models. Aircraft/frequency proxies could be assembled separately from operations, but have not been added to this specification; the other supplementary controls have not been assembled. True on-plane stops and plane changes cannot be recovered reliably from these files, and the coupon proxy replaces neither exactly.
- The original OLS load-factor coefficients are negative and the on-plane-stop coefficients are negative. Both modern load-factor and proxy coefficients are positive. Different controls, travel samples, proxy definitions and endogeneity prevent treating this as a like-for-like reversal of structural effects.
- This is 2025 Q2 DB1B with mileage-prorated directional Market fares, rather than 1987 Q3 preprocessed DB1A one-way/half-round-trip fares. Open-jaw/circle itineraries and first class cannot be excluded on the same basis as the historical study. The \$20 directional-fare floor is the documented pilot convention, not an inflation-adjusted historical cutoff; ancillary charges are not included.
- Carrier effects use the unchanged baseline's observed ticketing codes, including separate AS and HA categories. The parent-group combination used for the descriptive Q5 table is not imposed on this preselected regression baseline. Grouping differently would require rebuilding shares, fare percentiles and sample selection.

Inference and sampling limits. HC1 adjusts for heteroskedasticity, using $N/(N-K)$ times the robust covariance meat, but does not correct for endogeneity or correlation across routes sharing airports/networks. There is one observation per route, so route clustering would not address those shared-airport dependencies. No airport/network clustering is claimed. Standard errors treat constructed fare percentiles, shares and service measures as observed regressors/outcomes; they do not propagate all first-stage survey sampling uncertainty. Sparse carriers and the chosen market sample also limit generalization. Statistical significance is conditional on these assumptions.

Reproduction. Run the estimation and rendering scripts listed in the accompanying methodology, then compile the generated LaTeX with pdflatex. The output directory contains full coefficients (including carrier effects and the intercept), confidence intervals, fit statistics, effect conversions, exact sample rows and residuals. The source baseline SHA-256 matches its Q4 provenance; no baseline values changed. The design is full rank, and coefficients and HC1 covariance were independently verified numerically.